

Question

A certain organic solvent can extract salicylic acid from water, given its distribution coefficient $K_D = 100$, and the dimerization equilibrium constant of salicylic acid $K = 50$. It is known that the $pK_{a1} = 2.98$, $pK_{a2} = 13.66$ of salicylic acid.

The following questions are asked:

- Calculate the distribution ratio D of salicylic acid extracted by an equal volume of the solvent in water when the total concentration of salicylic acid in the original water is 0.1mol/L at $\text{pH} = 1.0$. (The answer should be taken to three significant figures)
- Calculate the distribution ratio D of salicylic acid extracted by an equal volume of the solvent in water when the total concentration of salicylic acid in the original water is 0.1mol/L under human physiological conditions. (The answer should be taken to two significant figures)
- Calculate the pH of the aqueous system when the distribution ratio $D = 1$. (The answer should be taken to one significant figure)

Which of the following statements is correct (calculation error within 2% can be considered correct):

- All other options are incorrect
- The answer to problem a is greater than 400
- The answer to question a is less than 360.
- The answer to problem b is 0.0021
- The answer to problem b is 0.0051

F. The answer to problem c is 5.2.

G. The answer to problem c is 5.6

H. The answer to problem c is 5.4.

Answer

Correct Answer: **H**

Detailed Explanation

According to the second-level ionization constant of salicylic acid, the second-level ionization of salicylic acid is extremely weak and can be ignored. Therefore, it is denoted as HA .

CHECKPOINT

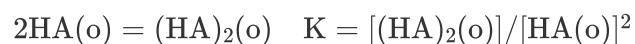
1 PTS

The second-level ionization of salicylic acid is extremely weak and can be ignored

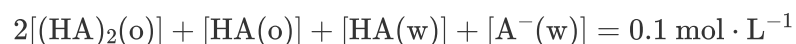
The suffix (w) represents species in water, and (o) represents species in organic solvent.

Problem a:

According to the equilibrium equation:



Equal volume extraction, so the material balance can be written as:



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Material balance $2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})] + [\text{HA}(\text{w})] + [\text{A}^-(\text{w})] = 0.1 \text{ mol} \cdot \text{L}^{-1}$

When $\text{pH} = 1$, we have $\delta_{\text{HA}} \approx 1$, ignoring $[\text{A}^-(\text{w})]$

CHECKPOINT

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When $\text{pH} = 1$, we have $\delta_{\text{HA}} \approx 1$, ignoring $[\text{A}^-(\text{w})]$

Rearranging and substituting gives:

$$2K[\text{HA}(\text{o})]^2 + (K_{\text{D}} + 1)[\text{HA}(\text{o})]/K_{\text{D}} - 0.1 = 0$$

Solving for

$$[\text{HA}(\text{o})] = 0.0270 \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{HA}(\text{w})] = 0.000270 \text{ mol} \cdot \text{L}^{-1}$$

$$[(\text{HA})_2(\text{o})] = 0.0364 \text{ mol} \cdot \text{L}^{-1}$$

$$D = (2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})])/[\text{HA}(\text{w})] = 370$$

CHECKPOINT

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$$D = (2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})]) / [\text{HA}(\text{w})] = 370$$

Problem b:

The pH of the human physiological environment is $\text{pH} = 7.4$. At this time, $\delta_{\text{A}^-} \approx 1$, ignoring $[\text{HA}(\text{w})]$.

CHECKPOINT

1 PTS

The pH of the human physiological environment is $\text{pH} = 7.4$. At this time, $\delta_{\text{A}^-} \approx 1$, ignoring $[\text{HA}(\text{w})]$.

Therefore, we have:

$$[\text{HA}(\text{o})] / [\text{A}^-(\text{w})] = K_{\text{D}}[\text{H}^+] / K_{\text{a1}} = 0.00380$$

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$$[\text{HA}(\text{o})] / [\text{A}^-(\text{w})] = K_{\text{D}}[\text{H}^+] / K_{\text{a1}} = 0.00380$$

Substituting into the material balance and rearranging gives

$$100[\text{HA}(\text{o})]^2 + 264[\text{HA}(\text{o})] - 0.1 = 0$$

Solving for

$$[\text{HA}(\text{o})] = 0.000379 \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{A}^-(\text{w})] = 0.10 \text{ mol} \cdot \text{L}^{-1}$$

$$[(\text{HA})_2(\text{o})] = 0.0000144 \text{ mol} \cdot \text{L}^{-1}$$

$$D = (2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})])/[\text{A}^-(\text{w})] = 0.0041$$

CHECKPOINT

1 PTS

$$D = (2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})])/[\text{A}^-(\text{w})] = 0.0041$$

Problem c:

$D = 1$, obviously pH is between 1 and 7.4, ignoring the second-level ionization.

Material balance: $2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})] = 0.05 \text{ mol} \cdot \text{L}^{-1}$; $[\text{HA}(\text{w})] + [\text{A}^-(\text{w})] = 0.05 \text{ mol} \cdot \text{L}^{-1}$

CHECKPOINT

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Material balance: $2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})] = 0.05 \text{ mol} \cdot \text{L}^{-1}$; $[\text{HA}(\text{w})] + [\text{A}^-(\text{w})] = 0.05 \text{ mol} \cdot \text{L}^{-1}$

Substituting the equilibrium constant and rearranging gives:

$$100[\text{HA}(\text{o})]^2 + [\text{HA}(\text{o})] - 0.05 = 0$$

$$[\text{HA}(\text{o})] = 0.0179 \text{ mol} \cdot \text{L}^{-1}$$

$$[(\text{HA})_2(\text{o})] = 0.016 \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{HA}(\text{w})] = 0.000179 \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{A}^-(\text{w})] = 0.05 \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{H}^+] = K_a[\text{HA}(\text{w})]/[\text{A}^-(\text{w})] = 3.75 \times 10^{-6} \text{ mol} \cdot \text{L}^{-1}$$

$$\text{pH} = 5.42$$

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$$\text{pH} = 5.42$$

Since the number of decimal places is calculated based on the significant figures of pH, retain one significant digit, which is 5.4.

Therefore, option H is correct.